Solar ESS Co. Ltd.
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INSTALLER GUIDE · BATTERY COMMUNICATION

Dyness: parallel battery communication

PowerBox Pro (10.24 kWh) wall battery · also applies to DL5.0C and A48100 / B4850 rack modules (CAN IN / CAN OUT)

One rule for every brand: only the **master** battery talks to the inverter. All other batteries are chained to it.

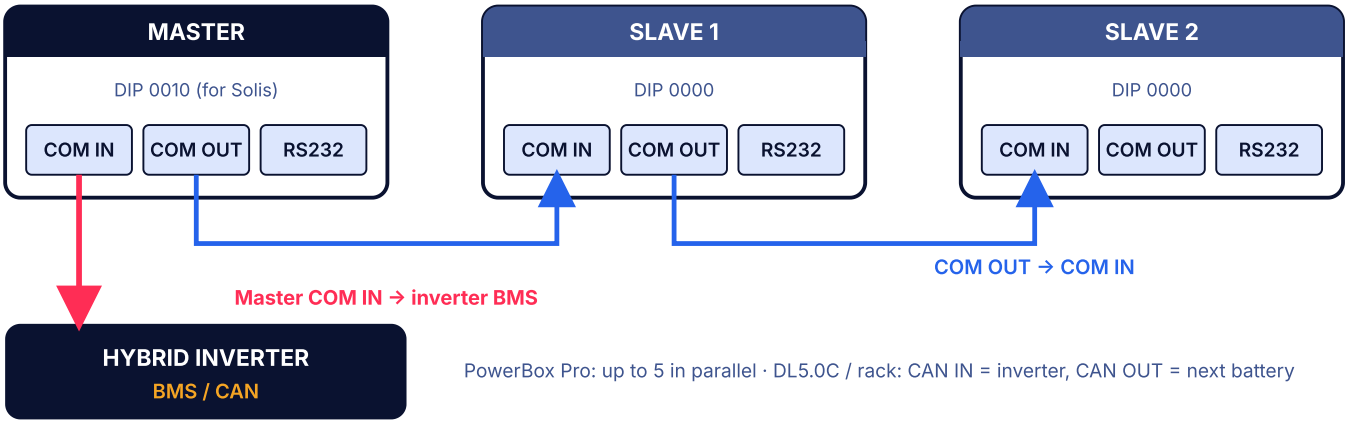
1 · Set the master / address



Master 0010	Master 0100	Master 0110	Every slave 0000
Solis, GoodWe, Deye, Sofar, Victron (CAN)	Growatt SPH / SPA (CAN)	Growatt SPF (RS485)	All OFF

Only the master is set, to match the inverter brand. Every slave stays at 0000 (all OFF). Switch 4 is baud rate: leave OFF (CAN 500k / RS485 9600). A single battery uses the master setting.

2 · Connect the communication cables



1. Inverter **BMS / CAN** → **Master COM IN** (on DL5.0C / rack modules: **CAN IN**).
2. **Master COM OUT** → **Slave 1 COM IN**, then Slave 1 COM OUT → Slave 2 COM IN, and so on.
3. The master's port pinout is CAN-H pin 4, CAN-L pin 5, matching Solis and Deye BMS ports. Use the cable labelled for the inverter if one is supplied.
4. On PowerBox Pro the manual places the master at the bottom of the stack.

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3 · Inverter settings

- **Solis:** Advanced Settings → Storage Energy Set → Battery Select → **Dyness LV**.
- **Deye:** Battery Mode → Lithium; BMS lithium protocol for Dyness/Pylon CAN (check the Deye manual for the number).
- **GoodWe:** select Dyness in the battery list (SolarGo app).
- **Growatt SPF:** Program 05 = LI, master DIP 0110 (RS485).
- Check: inverter max charge current ÷ one battery's current = number of batteries. If not, a slave is not communicating.

4 · Power-on sequence

1. DC breakers OFF. Set the master DIP, slaves 0000.
2. Switch on each battery (Reset / SW button). DL5.0C and rack modules: switch all on, then hold the master SW for 3 seconds.
3. Check all LEDs are green and the voltage at the inverter is above 44.8 V with correct polarity.
4. Close the DC breakers, then start the inverter.
5. Switch off: loads off, breakers off, then hold the master Reset for 3 seconds.

A 120 Ω CAN terminal resistor is supplied with PowerBox Pro. Fit it only where the manual for your unit shows it.

5 · Cable & good practice

Pinout	CAN-H pin 4, CAN-L pin 5, RS485-A pin 2, RS485-B pin 1.
Cable	Standard straight LAN cable (CAT5e/CAT6, T568B both ends). No crossover.
Routing	Keep communication cables away from DC power cables. Plugs must click in fully.
Mixing	Never mix brands, models or capacities in one battery bank or one communication chain.
Voltage	Before paralleling, charge or rest all packs so their voltages are close.

6 · Troubleshooting: "no SOC" or "BMS communication fail"

Master LED1 flashing yellow	Communication fault between batteries: check the COM OUT → COM IN cables.
LED1 and LED2 flashing alternately	Address fault: a slave DIP is not 0000, or a cable is in the wrong port.
Inverter shows no battery	Master DIP does not match the inverter brand, or the inverter cable is in a slave.
Battery never reaches 100%	Set the inverter charge voltage to 57.1–57.6 V (PowerBox Pro) or 56.5–57 V (DL5.0C).
Batteries switch off by themselves	Auto-sleep after about 72 hours without communication: check the inverter link.